CS-104 Quiz 2 (ANSWER KEY)

Fall 2026

Name: _____

This is an in-class, closed-book quiz (30 points). If you have any questions, please feel free to ask them. When you are finished, you can turn it in and go.

1. (2 points)

Fill in the blanks in each `range(...)` call so the loop prints exactly the numbers described (1 point each).

a) Every multiple of 6 from 6 to 60, inclusive:

```
for n in range(___, ___, ___):
    print(n)
```

b) The numbers 40, 36, 32, ..., 8, 4 (counting down by 4):

```
for n in range(___, ___, ___):
    print(n)
```

a) `range(6, 61, 6)` **(any stop from 61 to 66 works). b)** `range(40, 0, -4)` **(any stop from 0 to 3 works).**

2. (8 points)

Write a code segment that repeatedly inputs whole numbers from the user until the user enters `0`. Then it prints how many numbers were entered (not counting the 0), their average, and the smallest one. Do **not** use the built-in `min` function or a list. If no numbers were entered before the 0, print `no numbers` instead. Example run:

```
Number: 4
Number: 11
Number: -3
Number: 0
Count: 3, average: 4.0, smallest: -3
```

Just write a code segment; you don't need to include documentation, an algorithm, or test cases.

```
count = 0
total = 0
smallest = None
n = int(input("Number: "))
while n != 0:
    count = count + 1
    total = total + n
    if smallest is None or n < smallest:
        smallest = n
    n = int(input("Number: "))
if count == 0:
    print("no numbers")
else:
    print("Count:", count, "average:", total / count, "smallest:", smallest)
```

Grading: loop that stops at 0 2, count 1, average 1, smallest (works for positives) 2, "no numbers" case 1, output 1. Starting `smallest` at 0 loses the "smallest" point if all numbers are positive.

3. (4 points)

Consider the following (syntactically correct) code segment:

```
count = 0
for i in range(15, 0, -3):
    count = count + i
print(count)
```

a) What does it print? b) Write a `while` loop that does the same thing.

a) `45` (1 point). b) 3 points:

```
count = 0
i = 15
while i > 0:
    count = count + i
    i = i - 3
print(count)
```

4. (4 points)

What does this code print? Write every line of output.

```
total = 0
for i in range(2, 12, 3):
    if i % 2 == 0:
        total = total + i
    else:
        total = total - 1
    print(i, total)
print("done", total)
```

`2 2`, `5 1`, `8 9`, `11 8`, `done 8` **(grading: ¾ point per loop line, 1 point for the last line).**

5. (5 points)

Assume `x = 6`, `y = 11`, and `word = 'Bob'`. Give the value of each Python expression below (½ point each). If it raises an error instead, name the error.

```
>>> not False
```
True — example answer

```
>>> x > 5 and y < 11
```

```
>>> x > 5 or y < 11
```

```
>>> not x == 6
```

```
>>> 5 < x <= 6
```

```
>>> x % 2 == 0 and y % 2 == 0
```

```
>>> 'o' in word
```

```
>>> 'B' in word.lower()
```

```
>>> y == 6 or 7
```

```
>>> not (x > y)
```

```
>>> x + y > 15 and word == 'bob'
```

Answers: `False`, `True`, `False`, `True`, `False`, `True`, `False`, `7`, `True`, `False`.

6. (7 points)

Write a code segment that computes a movie ticket price. It inputs the customer's age (as an integer) and the day of the week (e.g. `Tue`). Then:

- If the age is negative, print `invalid age` and nothing else.
- Otherwise, the base price is $7 for ages under 12, $12 for ages 12 to 64, and $8 for ages 65 and up.
- On Tuesdays (`Tue`), every ticket is $3 off.
- Print the final price, e.g. `Price: 9`.

Just write a code segment; you don't need to include documentation, an algorithm, or test cases.

```
age = int(input("Age: "))
day = input("Day: ")
if age < 0:
    print("invalid age")
else:
    if age < 12:
        price = 7
    elif age <= 64:
        price = 12
    else:
        price = 8
    if day == "Tue":
        price = price - 3
    print("Price:", price)
```

Grading: inputs with conversion 1, invalid check 1, three age ranges 3, Tuesday discount 1, output 1.